

$$3d^4$$

July 23, 2026

$$1 \quad \left| 3d_{xz}^2 3d_{xy}^\alpha 3d_{x^2-y^2}^\alpha 3d_{yz}^0 3d_{z^2}^0 \right\rangle$$

Here are the CSFs:

$$\begin{aligned}
|1, 2, 2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|2, 2, 2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|3, 2, 2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|4, 2, 2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|5, 2, 2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|6, 2, 1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|7, 2, 1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|8, 2, 1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|9, 2, 1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

[illegible]

$$\begin{aligned}
|17, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|19, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|20, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|21, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|22, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|23, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|24, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|25, 2, -2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|26, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|27, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|28, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|29, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|30, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|31, 1, 0\rangle &= +\frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

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$$\begin{aligned}
|51, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|52, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|53, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|54, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|55, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|56, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|57, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|58, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|59, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

[illegible]

$$\begin{aligned}
|67, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|68, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|69, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|70, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|71, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|72, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|73, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|74, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|75, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|76, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|77, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|78, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|79, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|80, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|81, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|82, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|83, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|84, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|85, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|86, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|87, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|88, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|89, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|90, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|91, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|92, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|93, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|94, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|95, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|96, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|97, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|98, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|99, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|100, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|101, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|102, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

[illegible]

$$\begin{aligned}
|121, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|122, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|123, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|124, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|125, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|126, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|127, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|128, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|129, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|130, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|131, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|132, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|133, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|134, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|135, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|136, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|137, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|138, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|139, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle
\end{aligned}$$

$$\begin{aligned}
|140, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|141, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|142, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|143, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|144, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|145, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|146, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|147, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|148, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|149, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|150, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|151, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|152, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|153, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|154, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|155, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|156, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|157, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|158, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|159, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|160, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

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[illegible]

[illegible]

$$\begin{aligned}
|201, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|202, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|203, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|204, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|205, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|206, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|207, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|208, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|209, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|210, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 75 105 135

$$\begin{aligned}
\langle 75, 1, 1 | L_x | 71, 1, 1 \rangle &= -I \\
\langle 75, 1, 1 | L_x | 84, 1, 1 \rangle &= -I
\end{aligned}$$

$$\Delta g_{xx}$$

$$\begin{aligned}
&-\zeta \Delta_{75-71}^{-1} \\
&-\zeta \Delta_{75-84}^{-1}
\end{aligned}$$

$$\begin{aligned}
\langle 75, 1, 1 | L_y | 29, 1, 1 \rangle &= -\sqrt{\frac{3}{2}} I \\
\langle 75, 1, 1 | L_y | 44, 1, 1 \rangle &= -\frac{3}{\sqrt{2}} I \\
\langle 75, 1, 1 | L_y | 73, 1, 1 \rangle &= I \\
\langle 75, 1, 1 | L_y | 86, 1, 1 \rangle &= -I
\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}
&-\frac{3}{2} \zeta \Delta_{75-29}^{-1} \\
&-\frac{9}{2} \zeta \Delta_{75-44}^{-1} \\
&-\zeta \Delta_{75-73}^{-1}
\end{aligned}$$

$$-\zeta \Delta_{75-86}^{-1}$$

$$\langle 75, 1, 1 | L_z | 27, 1, 1 \rangle = -\sqrt{2}I$$

$$\Delta g_{zz}$$

$$-2\zeta \Delta_{75-27}^{-1}$$

For a multiplicity of 3 we have the following:

$$D = \langle 1, 1 | 1, 1 \rangle - \langle 1, 0 | 1, 0 \rangle$$

$$E = \langle 1, -1 | 1, 1 \rangle$$

$$\begin{aligned} \langle 1, 1 | \hat{H}_{eff} | 1, 1 \rangle = & -\zeta^2 (\\ & + \frac{3}{4} \Delta_4^{-1} + \frac{1}{4} \Delta_7^{-1} + \frac{1}{8} \Delta_{14}^{-1} + \frac{3}{16} \Delta_{34}^{-1} + \frac{1}{6} \Delta_{42}^{-1} + \frac{1}{16} \Delta_{49}^{-1} \\ & + \frac{1}{12} \Delta_{57}^{-1} + \frac{1}{8} \Delta_{64}^{-1} + \frac{1}{8} \Delta_{101}^{-1} + \frac{1}{8} \Delta_{103}^{-1} + \frac{1}{8} \Delta_{114}^{-1} + \frac{1}{8} \Delta_{116}^{-1} \\ & + \frac{3}{16} \Delta_{164}^{-1} + \frac{1}{16} \Delta_{169}^{-1} + \frac{1}{8} \Delta_{171}^{-1} + \frac{1}{8} \Delta_{173}^{-1} + \frac{1}{8} \Delta_{184}^{-1} + \frac{1}{8} \Delta_{186}^{-1} \\ &) \end{aligned}$$

$$\begin{aligned} \langle 1, 0 | \hat{H}_{eff} | 1, 0 \rangle = & -\zeta^2 (\\ & + \frac{3}{8} \Delta_9^{-1} + \frac{1}{3} \Delta_{12}^{-1} + \frac{3}{8} \Delta_{19}^{-1} + \frac{3}{16} \Delta_{29}^{-1} + \frac{3}{16} \Delta_{39}^{-1} + \frac{1}{16} \Delta_{44}^{-1} \\ & + \frac{1}{16} \Delta_{54}^{-1} + \frac{1}{8} \Delta_{59}^{-1} + \frac{1}{8} \Delta_{69}^{-1} + \frac{1}{8} \Delta_{71}^{-1} + \frac{1}{8} \Delta_{73}^{-1} + \frac{1}{8} \Delta_{84}^{-1} \\ & + \frac{1}{8} \Delta_{86}^{-1} + \frac{1}{8} \Delta_{131}^{-1} + \frac{1}{8} \Delta_{133}^{-1} + \frac{1}{8} \Delta_{144}^{-1} + \frac{1}{8} \Delta_{146}^{-1} + \frac{1}{6} \Delta_{167}^{-1} \\ & + 2\Delta_{202}^{-1} + 2\Delta_{204}^{-1}) \end{aligned}$$

$$\begin{aligned}
\langle 1, -1 | \hat{H}_{eff} | 1, 1 \rangle = & -\zeta^2 (\\
& + \frac{1}{8}\Delta_{14}^{-1} - \frac{3}{16}\Delta_{34}^{-1} - \frac{1}{16}\Delta_{49}^{-1} - \frac{1}{8}\Delta_{64}^{-1} + \frac{1}{8}\Delta_{101}^{-1} \\
& - \frac{1}{8}\Delta_{103}^{-1} + \frac{1}{8}\Delta_{114}^{-1} - \frac{1}{8}\Delta_{116}^{-1} + \frac{3}{16}\Delta_{164}^{-1} + \frac{1}{16}\Delta_{169}^{-1} \\
& - \frac{1}{8}\Delta_{171}^{-1} + \frac{1}{8}\Delta_{173}^{-1} - \frac{1}{8}\Delta_{184}^{-1} + \frac{1}{8}\Delta_{186}^{-1})
\end{aligned}$$

$$\begin{aligned}
D = & \zeta^2 (\\
& - \frac{3}{4}\Delta_4^{-1} - \frac{1}{4}\Delta_7^{-1} + \frac{3}{8}\Delta_9^{-1} + \frac{1}{3}\Delta_{12}^{-1} - \frac{1}{8}\Delta_{14}^{-1} + \frac{3}{8}\Delta_{19}^{-1} \\
& + \frac{3}{16}\Delta_{29}^{-1} - \frac{3}{16}\Delta_{34}^{-1} + \frac{3}{16}\Delta_{39}^{-1} - \frac{1}{6}\Delta_{42}^{-1} + \frac{1}{16}\Delta_{44}^{-1} - \frac{1}{16}\Delta_{49}^{-1} \\
& + \frac{1}{16}\Delta_{54}^{-1} - \frac{1}{12}\Delta_{57}^{-1} + \frac{1}{8}\Delta_{59}^{-1} - \frac{1}{8}\Delta_{64}^{-1} + \frac{1}{8}\Delta_{69}^{-1} + \frac{1}{8}\Delta_{71}^{-1} \\
& + \frac{1}{8}\Delta_{73}^{-1} + \frac{1}{8}\Delta_{84}^{-1} + \frac{1}{8}\Delta_{86}^{-1} - \frac{1}{8}\Delta_{101}^{-1} - \frac{1}{8}\Delta_{103}^{-1} - \frac{1}{8}\Delta_{114}^{-1} \\
& - \frac{1}{8}\Delta_{116}^{-1} + \frac{1}{8}\Delta_{131}^{-1} + \frac{1}{8}\Delta_{133}^{-1} + \frac{1}{8}\Delta_{144}^{-1} + \frac{1}{8}\Delta_{146}^{-1} - \frac{3}{16}\Delta_{164}^{-1} \\
& + \frac{1}{6}\Delta_{167}^{-1} - \frac{1}{16}\Delta_{169}^{-1} - \frac{1}{8}\Delta_{171}^{-1} - \frac{1}{8}\Delta_{173}^{-1} - \frac{1}{8}\Delta_{184}^{-1} - \frac{1}{8}\Delta_{186}^{-1} \\
& + 2\Delta_{202}^{-1} + 2\Delta_{204}^{-1})
\end{aligned}$$

$$\begin{aligned}
E = & \zeta^2 (\\
& - \frac{1}{8}\Delta_{14}^{-1} + \frac{3}{16}\Delta_{34}^{-1} + \frac{1}{16}\Delta_{49}^{-1} + \frac{1}{8}\Delta_{64}^{-1} - \frac{1}{8}\Delta_{101}^{-1} \\
& + \frac{1}{8}\Delta_{103}^{-1} - \frac{1}{8}\Delta_{114}^{-1} + \frac{1}{8}\Delta_{116}^{-1} - \frac{3}{16}\Delta_{164}^{-1} - \frac{1}{16}\Delta_{169}^{-1} \\
& + \frac{1}{8}\Delta_{171}^{-1} - \frac{1}{8}\Delta_{173}^{-1} + \frac{1}{8}\Delta_{184}^{-1} - \frac{1}{8}\Delta_{186}^{-1})
\end{aligned}$$